

force Characteristics, and Occupation-Specific Information.

Using O*NET’s task module, which contains 18,156 specific tasks, we describe each occupation as a vector of tasks typically performed by that role. We focus on tasks rather than skills or abilities, following the extensive “task approach” literature (Autor *et al.*, 2003, 2006; Acemoglu and Autor, 2011; Autor, 2013). This framework suggests that technological advancements displace or reinstate specific tasks within occupations rather than entire jobs or skills, as neither skills nor entire jobs directly translate into output (Autor, 2013).

To measure occupational exposure to AI, we employed a novel methodology using large language models (LLMs), specifically the GPT-4o API from OpenAI. Our approach, which we term “synthetic AI surveys,” involves asking these advanced models to estimate the likelihood of AI and its developments replacing each occupation based on its characteristic tasks.

Our synthetic AI survey methodology offers several key advantages in assessing occupational exposure to AI. First, the scalability of AI models allows for rapid processing of vast amounts of information, enabling a comprehensive analysis of all occupations in the database. This scalability ensures that no occupation is overlooked, providing a complete picture of potential AI impact across the labor market. Second, the use of a standardized AI model guarantees consistency in evaluation criteria across all occupations, eliminating potential human biases and ensuring comparability of results. Third, large language models like GPT-4o are trained on recent data, potentially incorporating the latest understanding of AI capabilities and trends, which keeps our analysis current and forward-looking. Lastly, these advanced models offer a more holistic view of tasks, occupations and technological innovations, capable of capturing interactions between tasks and considering broader contextual factors in their assessments.

Key features of our survey methodology include the following:

Independent Sessions: Each survey was conducted in a new session for every occupation, preventing carryover effects from previous responses. This ensures that each occupation is evaluated independently, based solely on its own characteristics.

Multiple Iterations: The process was repeated 100 times to test the accuracy and consistency of the estimates. This multiple-iteration approach allows us to account for potential variability in the AI model’s responses and provides a measure of confidence in our results.

Varied Creativity: The temperature parameter, which controls the randomness and creativity in the model’s output, was randomly set between 0 and 2 for each iteration. This introduces controlled variability in the model’s responses, potentially capturing a wider range of perspectives on each occupation’s AI exposure.

Time Horizons: We considered three exposure horizons: one year, five years, and 10 years. This allows us to capture both immediate and longer-term potential impacts of AI on occupations, recognizing that AI capabilities and their integration into various fields may evolve over time.

The GENOE index for each horizon was calculated as the average value across the 100 sur-